



Whitepaper: Web Service Architecture Tenets

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Web Service Architecture Tenets

Software construction and integration continues to be a labor intensive task. As the state of computer science progresses, we are continually finding new tools and techniques to reduce the efforts and increase the return. The rise of a ubiquitous service oriented computing theme has brought new opportunity for productivity and new challenges for architects and designers. The problems and solutions can be organized into a few overarching groups:

1. Working with services units (focus on loose coupling)
2. Working with networks (focus on resource virtualization)
3. Networking services together (focus on agile composition)
4. Doing 1-3 economically (focus on scalability across several dimensions)
5. Enabling new offerings based on the new technology (focus on the future)

1. Consumers and Producers are Loosely Coupled

Loosely coupling software assets is of primary importance if they are to be reusable and integrate-able. Attempts to enable reuse in platform specific, object oriented scenarios have failed. Modern WSA (Web Service Architecture) programs exist primarily to provide an atmosphere where functionality can be reused. Loose coupling can:

A. Provide Functional Encapsulation

Functional encapsulation is the process of hiding the internal workings of a service to the outside world. This is a concept that was highly promoted in the object oriented world and continues to be a primary consideration in designing a service.

B. Factor Out Non-Functional Concerns

One of the primary impediments to reuse has been that the remedies to non-functional concerns effectively “glued” components together (i.e. tight coupling). Common examples of these concerns included reliability, security and integrity, and the corresponding remedies included: reliable messaging, triple-DES encryption, and atomic transactions. Unfortunately, references to these remedies were often hard coded into our components, impeding their encapsulation, exposure, and ultimate reuse. In contrast, the SOA model encourages the use of standardized policies that advertise the remedies to these non-functional concerns. In this model, a consumer can “read” the policy and determine if it will serve its needs. If so, the consumer can move forward in using the service. If not, the consumer can query the network to determine if a mediating device is available to provide the necessary translation.

C. Force Ubiquity at the Edge of the Service

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The Web Services umbrella is fundamentally a collection of standards. SOAP, WSDL, and the WS-* specifications are designed to standardize any non-functional application aspect (i.e. anything that IS NOT functionally encapsulated). The goal of these standards is to provide ubiquity in the programming model at the edge of the service. Introducing a new protocol or format that lives on the outside of the service but can't be recognized or consumed will reduce the reusability of the service and force a tight coupling between the intended consumer and actual producer.

2. Leverage Network Computing and Resource Virtualization

In the 1990's, a significant focus was placed on virtualizing the machine, an effort that produced the "virtual machine", or VM. The web service computing paradigm takes this concept one step farther – now the emphasis is on virtualizing many machines across a network. This "virtual network of machines" is usually referred to as a 'grid'. Although the web services architecture doesn't mandate full resource virtualization, it does offer a stepping stone to this capability. This is attractive to organizations that are interested in implementing autonomic or on-demand operational models. Virtualization can:

A. Network the Services

As services are created, their interfaces and policies are advertised using ubiquitous formats. This advertisement of capability becomes the software contract. As more and more contracts are created and placed on the network, a Service Network is created. The services along with intermediaries become the basic building blocks for creating and integrating software. In network computing, a single programming style is often used for both the 'local network' (the local computer) and the 'remote network'. (This is the virtualization of the network itself).

B. Provide Mediation in the Network

SOA enables a 'network programming model'. This allows intermediaries or 'bumps in the network' to provide value added functions to the messages that are traveling between producers and consumers. Mediating functions include: message transformation, content based routing, load balancing, authentication and many others. These intermediaries not only help in the decoupling effort, but they can also be used as remedies to non-functional concerns like availability and integrity.

C. Resolve Resources at Runtime

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Resource oriented computing focuses on the virtualization of computing resources. This means that special attention is placed on discovering resources at runtime and using those resources in a fashion whereby they can be released and regained.

3. Systems are Loosely Bound in the Large and Small

The use of web service contracts (interface + policy) enable systems to be loosely bound for both the composition of systems as well as the integration of those systems. With web service contracts:

A. Integration and Composition become One

Historically, software composition and software integration leveraged different programmatic mechanisms. Software was traditionally composed by using objects and language-specific contracts or interfaces; applications were traditionally integrated using proprietary messaging systems. As messaging and interface standards become ubiquitous, we begin to see a single unified model that embraces standardized messaging across standardized interfaces. This use of a unified programming model blurs the distinction between integration and composition. The remaining distinction becomes a matter of programmability. Current compilers and platforms are tuned to the language specific composition problem. Work is still needed to make the programmability of a single model as easy as the current state of the art in the object oriented world.

B. Functionally Structured, Operationally Amorphous

In the WSA, sub-system functionality is exposed as a set of contracts. These contracts are bound through architectural and compositional patterns (runtime location, correlation, leasing, eventing, etc.) which are implemented in the language of choice and are executed in an operational location that may be determined at runtime.

B.1 Enterprises Architect the Manageability of their Critical Service Network

Systems that are deemed critical to the enterprise are ‘designed-for-manageability’. That is, those systems, along with the services and intermediaries upon which they depend, are checked to verify that the entire loosely bound system will work as designed. The compilation of loosely bound systems is significantly more important in mature networks where services are shared and new versions of shared services are introduced.

B.2 Non-Critical Service Networks Grow Organically

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Systems that are deemed non-critical can easily be introduced and removed from the service network without a significant negative impact. Metcalfe's law states that, "*the usefulness, or utility, of a network equals the square of the number of users.*" A variation of this law is applicable to the Service Network, that is, "*the usefulness, or utility, of a services network equals the square of the sum of the number of easily consumable and valuable services available on the network.*" A significant amount of the services available on a network will be simple, non-critical, user centric services that may be managed organically.

4. Enterprise Service Networks Scale Efficiently



Unlike many other computing styles, the service network is meant to scale. It should facilitate hundreds or thousands of different applications. Object oriented, RAD and platform specific systems faced difficulty in scaling the number of applications due to reuse of a common infrastructure. Packaged Applications faced scaling obstacles with regard to extensibility, configurability and integrate-ability. The web service architecture incurs additional upfront burden to facilitate massive scaling down the road. An Enterprise Service Network:

A. Contains a Managed Enterprise Vocabulary

The primary vehicle for a programmer is his language. Teams of programmers extend their languages by adding functions (or services). These services introduce terms that closely resemble the task at hand (e.g., shipOrder, bookFlight). Enterprises create vocabularies of nouns (Flight, Order, etc.), verbs (ship, book, etc.) adjectives (Flight Number) and now adverbs (reliably, securely, etc.) Corporations have placed significant attention on managing their nouns and adjectives (mostly for their database systems), but have placed virtually no emphasis on managing their verbs or adverbs. As the enterprise shifts into a service oriented computing paradigm, managing the verbs and the adverbs will become as important as managing the nouns and adjectives was in the last generation. Also, the managing of the nouns will need to span more partitions, for example: 1. Service Messages, 2. Objects, 3. Databases and 4. Specialized Containers (e.g., rules engines, orchestration engines, etc.). The creation of a common enterprise vocabulary allows distributed departments to leverage prior works and build out a common semantic and functional vocabulary.

B. Facilitates Service Use and Reuse

If a programmer can't find a service, they won't use it. If they find it, and it doesn't fit their needs, they won't use it. If it does fit their needs, but they have to jump through hoops to incorporate it, they won't use it. As you build out your service network, make it easy for developers to find and use the services. Conversely, make it

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easy for the developer to contribute their services back to the shared pool. The intent of the code should be self revealing or explicit, enabling asset harvesting.

C. Creates a Federated Solution

The Service Network conjoins a number of traditional enterprise computing boundaries. This includes network and administrative domain boundaries, organizational and operational boundaries as well as the boundaries of time and space. The capabilities of – and services in – the service network will evolve and mature with independence. Nonetheless, cohesion across the nodes in the service network is advantageous in order to minimize impedances. Balancing these concerns, and their associated remedies, requires that control and evolution of the capabilities (and associated services) in the service network be federated. A closer look at this dynamic bears out an interesting conclusion. The capabilities of – and relationships between – services in the service network are inherently symbiotic. The dynamics of this relationship can be summarized as a form of inversion dependency – all of the capabilities of and relationships between services in a service network are nodes on the network, and all service networks are comprised of service nodes. The two are interdependent – each depends on the other's existence and capabilities, and neither exists without the other. Therefore, the services in the service network are in no way autonomous – rather they comprise a federation of service nodes with varying capabilities and relationships to each other that evolves symbiotically across enterprise computing boundaries, time and space.

5. WSA Enables New Opportunities

In each major paradigm shift, we've seen that technology was able to easily solve a business problem. The advent of the WSA is no exception. At the core of the WSA promise is the ability to share information – between people, systems and corporations. With a WSA:

A. Enterprise Boundaries are Removed

The Internet and the Web demolished enterprise boundaries. By deploying ubiquitous protocols, information systems quickly became geared toward communities of interest, regardless of where the corporate boundaries were drawn. Web services hold the potential to further destroy 'intranet thinking'. The ability to quickly share information between consumers and retailers, distributors, manufacturers, professional service providers, and other members of the value chain, enables just-in-time decision making that is optimized across boundaries. The efficiencies gained by 'self-service applications' will be extended past the single corporation. The concept

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of “one face to the customer” will now be available across corporations and consumers will likely come to demand it.

B. Unstructured Data Finds Structure and Meaning

The activities associated with moving towards service network overlays, managed enterprise vocabularies and enterprise-scope metadata repositories has a natural set of consequences. Information models in this environment are inherently descriptive. We see standardization on schema-based grammars for defining, modeling and advertising the artifacts of the service network. As a result, we are beginning to organically remove many of the semantic impedances that have traditionally impacted the drive towards ubiquitous, business-driven information and process integration. By using schemas and semantic knowledge, we are further enabling the power users to find the natural pivot points that exist in their day-to-day information systems.

C. Systems are Tolerant to Change

Enterprises implement a wide variety of enterprise information systems that support different aspects of planning, execution, monitoring and analyzing business processes. Some of these systems take the form of packaged enterprise applications, while others are custom applications. Yet others were at some point packaged and through time have been extended and modified to the point that for all intents-and-purposes they are now fully customized applications. For the most part these systems were designed, developed and deployed under the influence of the IT-based “initiative-du-jour” which was synonymous with the respective era of computing.

Examples of these initiatives include data warehousing and OSS, ERP, CRM and SCM. These initiatives evolved and matured over time from stand-alone, custom applications (developed and integrated using proprietary languages and interfaces; designed with inherently centralized architectures, and deployed onto mainframe-based computing platforms); to networked, packaged applications (developed and integrated using standardized languages and interfaces; designed with inherently-decentralized architectures and deployed onto commodity-based computing platforms). Therefore, the IT-driven responsiveness, required by enterprise-scope business initiatives, has been inordinately constrained by these “bottom-up”, language, platform and API-specific approaches to evolving these systems.

As a result, enterprises have viewed the art (or voodoo) of evolving these systems as the exclusive domain of “rocket scientists”. Accordingly, IT as whole – and in particular the systems infrastructure supporting enterprise initiatives and their corresponding business process – have been mostly misaligned with the operational goals and strategic objectives driving the business. With a move toward design-by-contract, contract-first-construction and protocol-based interoperability and

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integration, there is an opportunity to enable tolerant evolutions through malleable abstractions and mediate impedances at run-time.

D. Power Users can Impact the System

As is the case with the other phases of enterprise computing, business-driven, service-oriented computing is beginning to mature and gain strategic enterprise adoption and applicability. This maturation is driving the next natural set of evolutions, forming preferred architectural approaches, patterns, methods, processes and roles. As service fabric overlays, enterprise-scope application-oriented service networking and managed service networks rapidly enter the mainstream, we are seeing that the traditional notions applications and platforms are being recast around “network application platforms”. We are moving the “center of gravity” from server-based containers, registries and consoles, to network-centric, run-time intermediaries, metadata management services, and goal-directed interaction interfaces.

As the network becomes smarter, the abstractions and interaction mechanisms become closer to the capabilities of the average business worker. Just as employees became an integral part of Intranets by feeding and consuming information, a similar dynamic is beginning to occur with the service network. Power users are able to access information through desktop productivity tools and are able to easily share the information and collaborate with their co-workers.

Concluding Remarks

The transition to a Web Service Architecture is more significant than “just using SOAP” or adopting distributed computing. If properly applied, WSA has the potential to provide a sustainable competitive advantage to any corporation that successfully adopts it – and poses an equally daunting fate for those organizations that fail to adopt it.

About the Authors

Jeff Schneider is founder and CEO of MomentumSI, a consulting firm that specializes in transforming organizations into a “Service Oriented Enterprise”.

Frank Martinez is founder and CTO of Blue Titan Software, the services networking company. Blue Titan’s products help enterprises deploy and manage their service-oriented architectures by enabling IT to control, share and scale services and applications. Business innovation can therefore be driven across the distributed enterprise in a timely, cost effective and future-ready manner.

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